OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

12.13 Junction groups

Two or more junctions may be grouped in junction groups to indicate that these junctions belong to the same roundabout.

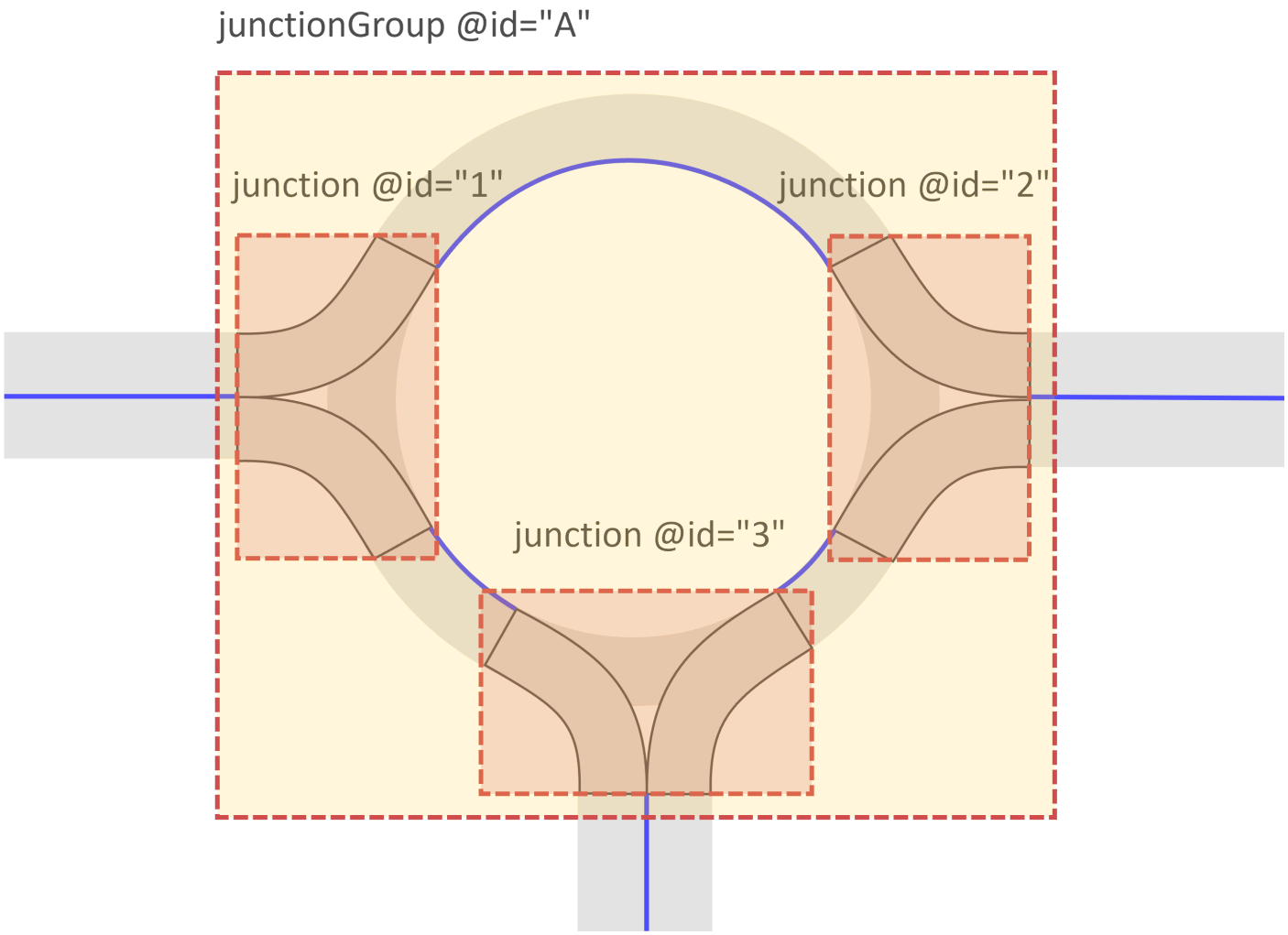

Figure 105. Junction group with three junctions

Figure 105 shows how the junctions `1`, `2` and `3` are aggregated in junction group `A`.

Junction groups are described by `<junctionGroup>` elements. The junctions that belong to the junction group are specified by `<junctionReference>` elements.

Elements in UML model

`<junctionGroup>` element

In ASAM OpenDRIVE, junction groups are represented by the `<junctionGroup>` element within the `<OpenDRIVE>` element.

```
UML class: t_junctionGroup
XML tag:   <junctionGroup> (Multiplicity: 0..*)
```

Junction groups indicate for routing that the grouped junctions belong to the same node and are commonly seen as one big junction, for example roundabouts or highway interchanges.

The `<junctionGroup>` element is split into a header element and a series of member elements.

Table 83. Attributes of the <junctionGroup> element

Name	Type	Use	Description
`id`	string	required	Unique ID within database
`name`	string	optional	Name of the junction group. May be chosen freely.
`type`	e_junctionGroup_type	required	Type of junction group

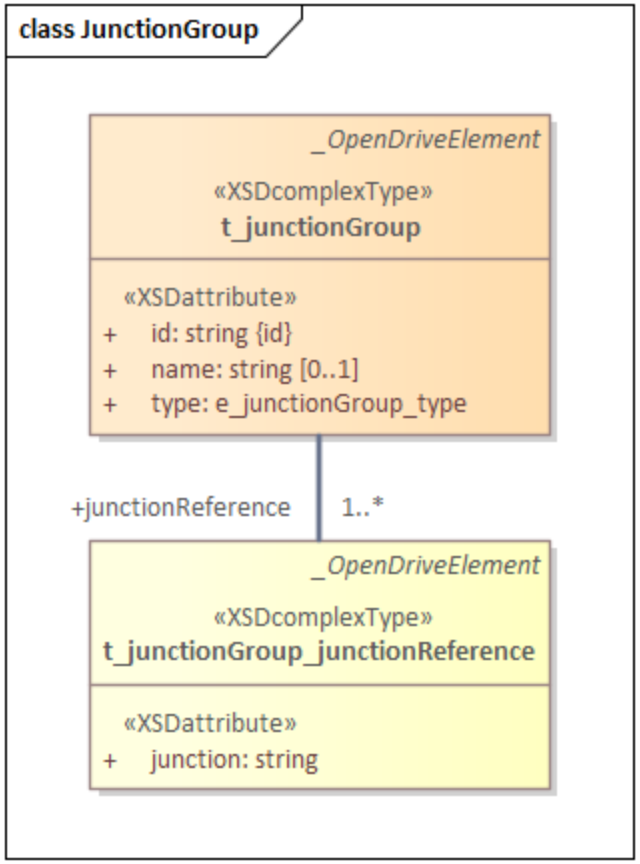

Figure 106. UML class diagram of the JunctionGroup class

Figure 106 shows the UML class diagram of the ASAM OpenDRIVE JunctionGroup class.

`<junctionReference>` element

In ASAM OpenDRIVE, references to junctions are represented by the `<junctionReference>` element within the `<junctionGroup>` element.

```
UML class: t_junctionGroup_junctionReference
XML tag:   <junctionReference> (Multiplicity: 1..*)
```

References to existing `<junction>` elements.

Table 84. Attributes of the <junctionReference> element

Name	Type	Use	Description
`junction`	string	required	ID of the junction

XML example

- UC_2Lane-RoundAbout-3Arms.xodr

Related topics

- Section 12.1, "Introduction to junctions"
- Section 12.14, "Signal synchronization groups in junctions"